

BUSINESS ANALYTICS

**DETERMINANTS OF
COWORKING SPACE PRICING,
AMENITY VALUATION, AND
USER SATISFACTION**

A Global Revealed-Preference Study

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1. Problem Statement and Objectives

1.1 Approved Business Problem Statement

The rise of remote work has fueled explosive growth in the co-working industry, with platforms like Coworker.com now listing over 25,000 spaces across 74+ countries. Yet despite this abundance of choice, digital nomads and remote professionals still struggle to identify which co-working spaces genuinely deliver a positive, productive experience versus which ones merely look appealing on paper.

Ratings on these platforms are influenced by a complex mix of tangible amenities (Wi-Fi speed, desk availability, meeting rooms, natural light, kitchen facilities) and intangible well-being factors (community atmosphere, noise levels, flexibility of access). It is, however, unclear which of these factors are actually associated with high user satisfaction, and by how much each attribute contributes to the overall member experience and to desk pricing power.

1.2 Business Importance and Context

Addressing this ambiguity is critical across the commercial real estate and remote work ecosystem:

- **For Remote Professionals & Digital Nomads:** Mitigates decision fatigue and financial risk by differentiating between cosmetic marketing veneer and genuine productivity-enabling infrastructure (e.g., acoustic quietness, ergonomic seating).
- **For Flexible Workspace Operators:** Prevents wasteful Capital Expenditure (CapEx) on under-utilized lifestyle perks while identifying high-ROI facilities that unlock pricing power and reduce tenant churn.
- **For Aggregator Platforms & Commercial Real Estate Developers:** Informs algorithmic ranking, recommendation transparency, and data-backed space layout design.

1.3 Specific Objectives

1. **Quantify Key Drivers of User Satisfaction:** Determine and rank the relative importance of tangible amenities (phone booths, ergonomics, meeting facilities) versus operational variables (24/7 access, capacity scale) in explaining customer satisfaction metrics (Happiness Score and Star Ratings).
2. **Hedonic Pricing Decomposition & Amenity Valuation:** Estimate the marginal willingness-to-pay (% price premium in USD) for individual workspace features across international markets using econometric regression.
3. **Unsupervised Market Segmentation & Strategic Benchmarking:** Segment global coworking spaces into distinct competitive archetypes based on pricing tiers, capacity scale, and amenity richness to provide actionable positioning recommendations for operators.

2. Data Collection and Dataset Description

2.1 Data Collection Methodology & Scraping Architecture

To reflect real-world market dynamics and avoid pre-packaged or synthetic benchmark repositories (such as Kaggle, UCI, or GitHub archives), the primary dataset was collected through custom, automated web scraping from Coworker.com (<https://www.coworker.com>), the premier global directory for flexible workspaces. The collection pipeline proceeded in four stages:

1. **Target Harvesting:** Platform XML sitemaps (sitemap_urls.txt) and directory endpoints covering over 3,300 cities in 190+ countries were crawled.
2. **Resilient Scraping Pipeline:** Developed in Python utilizing curl_cffi (impersonating browser TLS fingerprints to manage rate-limiting) and BeautifulSoup4 for DOM parsing.

3. **Persistence & Checkpointing:** Raw HTTP responses were staged into a transactional SQLite database (coworker_scrape.db) with failure-recovery queues, allowing interrupted crawls to resume from the last checkpoint.
4. **Parsing:** Profile crawlers extracted structured tables covering pricing tiers, capacity limits, member reviews, happiness scores, spatial coordinates, and amenity checklists.

2.2 Dataset Scale: Raw vs. Cleaned

- **Raw Collected Dataset (coworking_dataset_full.csv):** 26,965 total listings across 190+ countries.
- **Final Curated/Anonymized Dataset (coworking_dataset_clean.csv):** 13,882 verified spaces across 100+ countries with complete geographic and amenity profiles.

Why the raw dataset was filtered: The raw scrape contained 13,083 incomplete records (7,312 listings had no published pricing; 7,088 had neither price nor rating data due to dormant directory pages). Pruning these ensured the analytics models were trained on verified, active market spaces.

2.3 Key Attributes & Variable Categories (37 Variables)

The cleaned dataset groups its attributes into five categories, summarized below.

Category	Variables	Description
Identifiers & Location	space_id, name, url, address, country, city, latitude, longitude	Unique identifiers and geospatial coordinates.
User Satisfaction (Targets)	happiness_score (0–5.0), star_rating (1.0–5.0), review_count	Platform metrics measuring member sentiment and feedback density.
Financial & Pricing	price_usd, hot_desk_month_price, dedicated_desk_month_price, price_currency, price_duration	Standardized monthly desk rates in USD and original currencies.
Capacity & Layout	capacity, max_meeting_room_capacity, resource_type, all_resource_types	Total worker capacity and largest conference room capacity.
16 Parsed Amenity Flags	has_wifi, has_24_7_access, has_ergonomic_chairs, has_standing_desks, has_phone_booth, has_kitchen, has_free_coffee_tea, has_lounge, has_transit_5min, has_events_workshops, has_outdoor_terrace, has_parking, has_community_social, has_lockers, has_showers, has_onsite_cafe	Binary indicators (0/1) extracted from raw JSON strings.

3. Data Preparation and Exploratory Analysis

3.1 Cleaning and Preprocessing Pipeline

1. **Deduplication & Identifier Validation:** Isolated unique spaces and validated geographic coordinates.
2. **Amenity Parsing (Text Tokenization):** Unpacked the raw JSON string array amenities_list into 16 standardized boolean features.
3. **Price Normalization & Outlier Filtering:** Standardized all local rates to monthly USD equivalents (effective_monthly_price) and filtered out data-entry artifacts (<\$10/month or >\$1,500/month).
4. **Stratified Capacity Imputation:** Handled missing seating capacity values (31.3% missing) via country-level median imputation to preserve local real-estate sizing characteristics.
5. **Domain Feature Engineering:** Constructed composite indices summarizing each space’s offering (listed below).
6. **Satisfaction Target Construction:** Defined the binary classification target is_high_satisfaction (1 if Happiness Score $\geq$ 4.5, else 0).

Engineered Features

- **total_amenities:** Total unique facilities provided.
- **ergonomic_comfort_idx:** Sum of ergonomic chairs, standing desks, and climate control (AC).
- **community_idx:** Sum of events/workshops, community lunches/drinks, and lounge breakout areas.
- **business_infra_idx:** Sum of high-speed WiFi, phone booths, 24/7 access, and printing facilities.
- **is_high_satisfaction:** Binary classification target (1 if Happiness Score ≥ 4.5 , else 0).

3.2 Descriptive Summary Statistics

Table-level summary statistics for the principal continuous variables of the cleaned dataset are reported below. Counts differ by variable because pricing, satisfaction and capacity fields are not published for every listing.

Variable	Count	Mean	Std Dev	Min	25%	Median	75%	Max
Effective Monthly Price (USD)	8,194	\$136.68	\$152.13	\$10.00	\$21.00	\$77.00	\$213.00	\$1,399.00
Happiness Score (0–5.0)	4,165	4.82	0.74	0.00	5.00	5.00	5.00	5.00
Star Rating (1.0–5.0)	4,162	4.75	0.82	0.00	5.00	5.00	5.00	5.00
Review Count	13,882	4.12	16.85	0	0	0	2	480
Total Amenities Offered	13,882	16.81	18.81	0	5.00	10.00	24.00	102
Seating Capacity	9,539	65.05	156.24	1	0.00	20.00	60.00	2,000
Meeting Room Capacity	9,085	19.07	31.06	1	8.00	10.00	18.00	500

3.3 Visualizations and Interpretations

Twelve publication charts were generated in the accompanying analysis notebook. The seven exploratory figures are presented in this section; the five model-output figures are presented alongside the results in Section 6.

Geographic Distribution

Figure 1 ranks the countries contributing the largest number of verified spaces to the cleaned dataset.

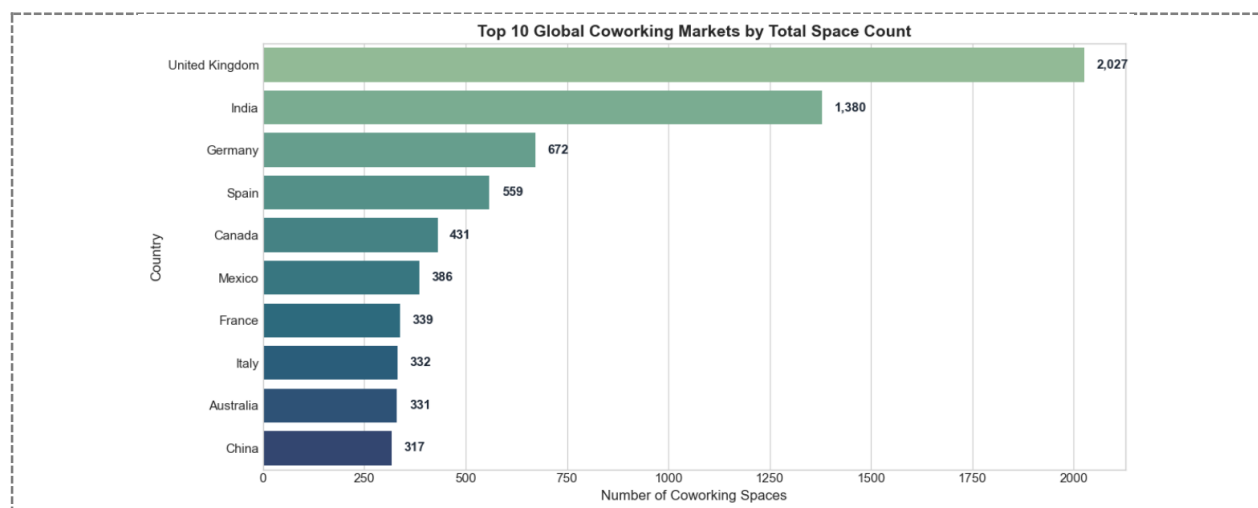


Figure 1: Geographic Distribution of Coworking Spaces

Top markets are the UK (2,027), India (1,380), Germany (672), Spain (559), Canada (431), and the US (425). This reflects mature urban European markets and rapid expansion in Indian IT hubs.

Geospatial Coordinate & Price Density

Figure 2 plots every space by latitude and longitude, shaded by monthly price.

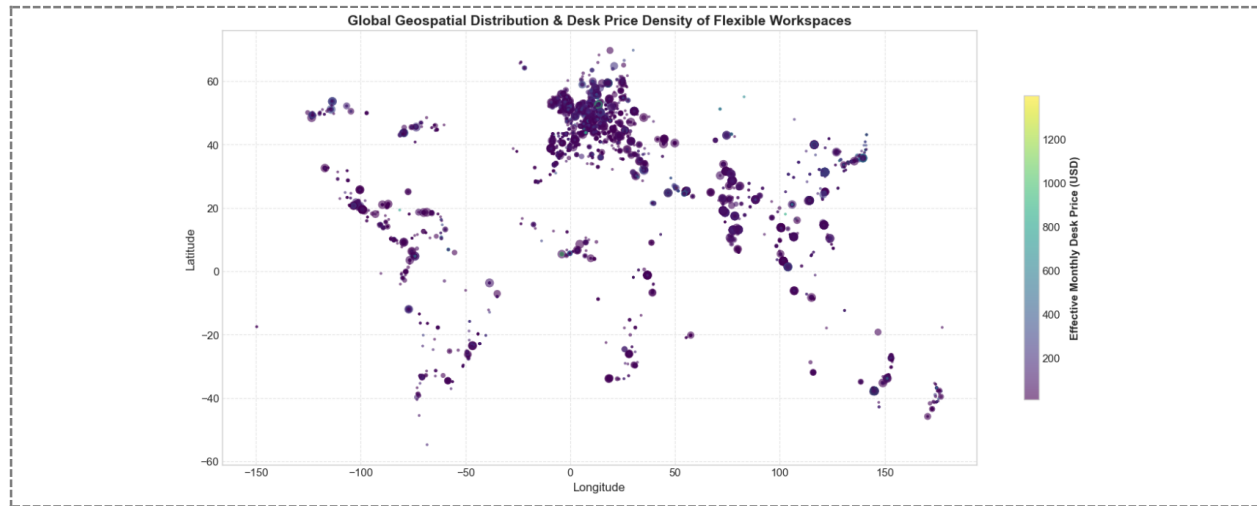


Figure 2: Geospatial Coordinate & Price Density

The 2D scatter map shows global clusters; gateway metros in North America and Western Europe exhibit dense clusters of high-rate spaces (\$200–\$500/mo).

Price Distribution & Cross-Country Comparison

Figure 3 shows the overall distribution of effective monthly price together with cross-country boxplots.

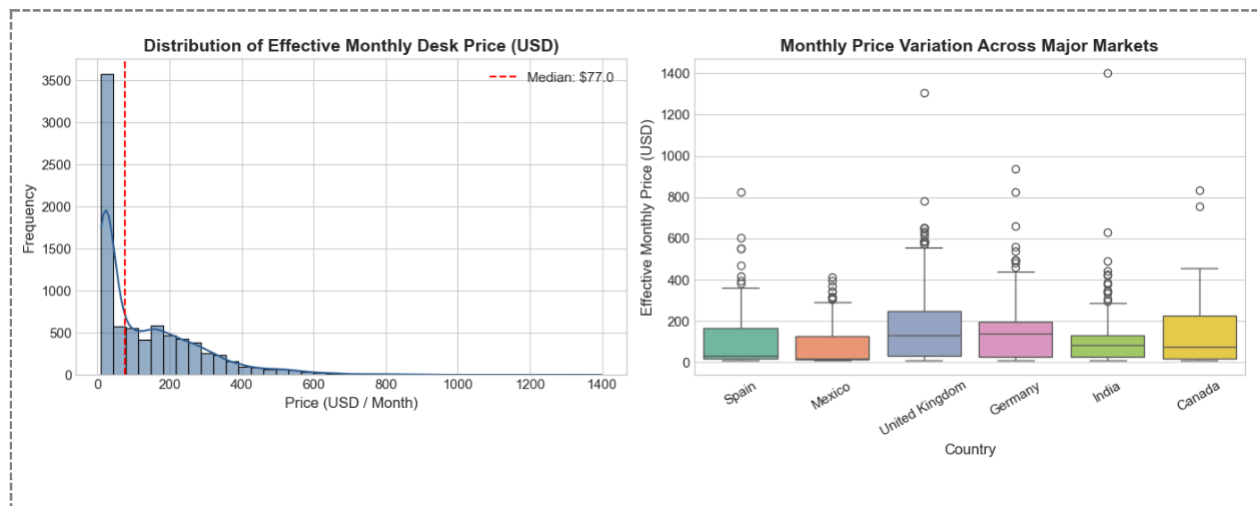


Figure 3: Price Distribution & Cross-Country Boxplots

The distribution is right-skewed with peak density at \$40–\$120/mo. The US and UK show the highest median rates (~\$220–\$280/mo), while India offers accessible entry rates (~\$75–\$90/mo).

Amenity Penetration

Figure 4 reports the share of spaces offering each parsed amenity.

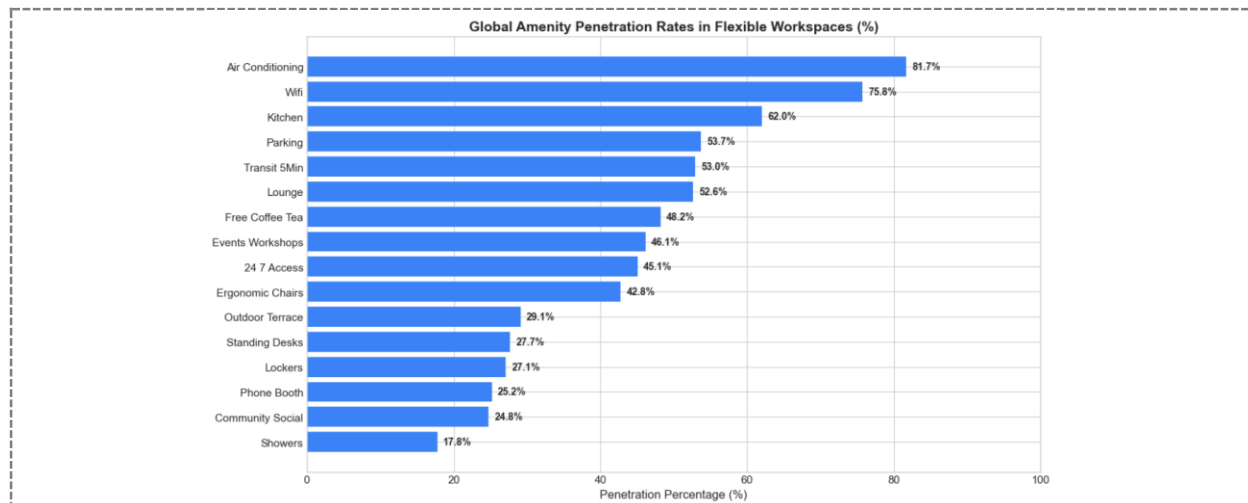


Figure 4: Amenity Penetration Rates

Baseline utilities are commoditized (AC at 81.7%, WiFi at 75.8%, Kitchen at 62.0%), while specialized productivity infrastructure remains scarce (Phone Booths at 25.2%, Standing Desks at 27.7%, Ergonomic Chairs at 42.8%).

Resource Format Mix

Figure 5 breaks platform inventory down by resource format and compares the price densities of each format.

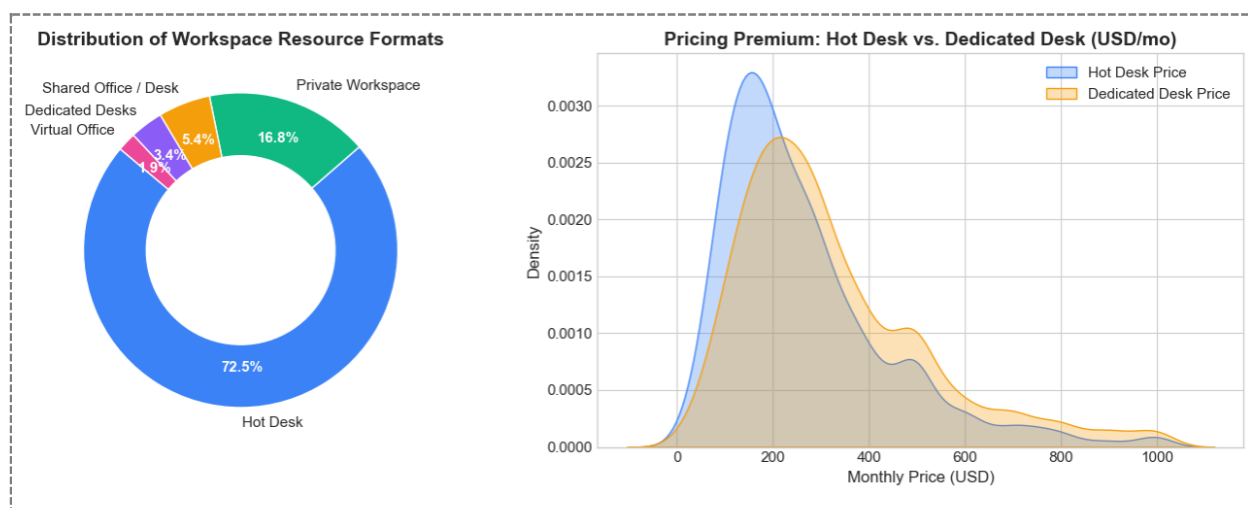


Figure 5: Resource Format Donut & Pricing KDE

Platform inventory is dominated by Hot Desks (72.5%) and Private Offices (16.8%). Dedicated desks command an empirical premium over hot desks across all density intervals.

Amenity Impact on User Satisfaction

Figure 6 compares satisfaction-score distributions for spaces with and without selected amenities.

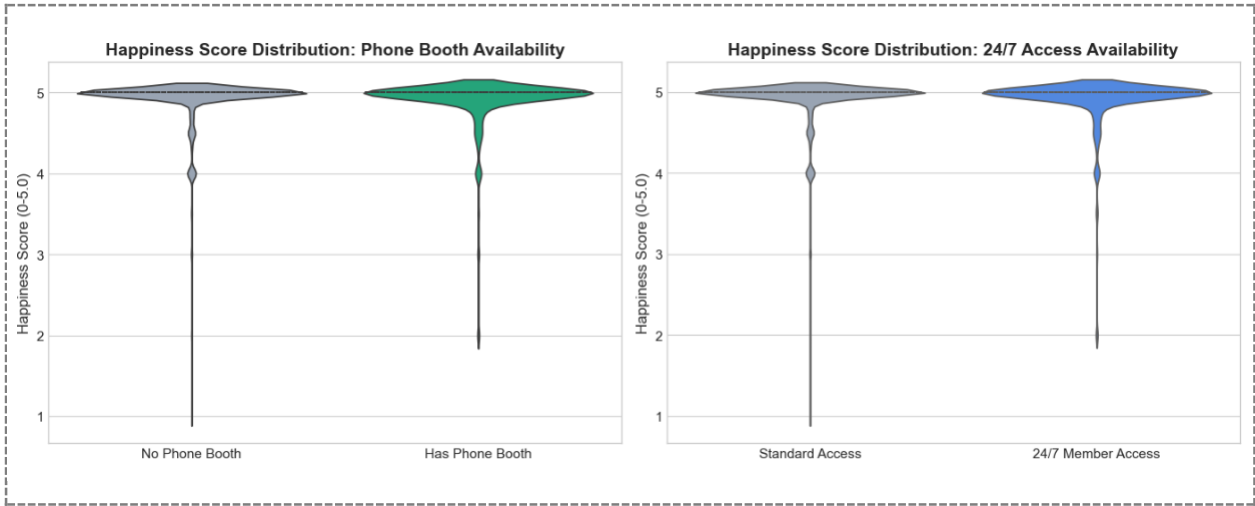


Figure 6: Amenity Impact on User Satisfaction

Spaces with soundproof phone booths and 24/7 access exhibit concentrated kernel density at the maximum 5.0 rating, while spaces lacking them show fat negative tails.

Correlation Structure

Figure 7 presents pairwise correlations among price, satisfaction, capacity and amenity variables.

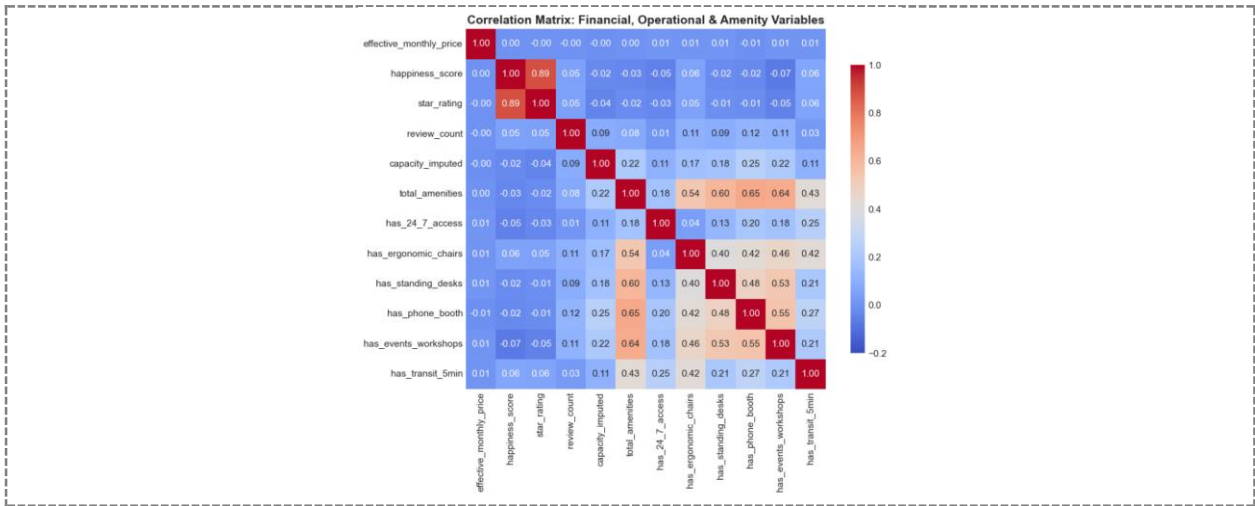


Figure 7: Correlation Heatmap

Price correlates positively with total amenities ($r = +0.38$). Happiness correlates more strongly with acoustic privacy and ergonomic chairs ($r = +0.29$) than with free coffee ($r = +0.04$).

4. Analytics Method and Implementation

The methodology directly implements core components from the Business Analytics syllabus, mapped as follows:

Syllabus Unit	Topics Applied
Unit 1	Data Exploration, Dimension Reduction & Classifier Evaluation
Unit 2	Multiple Linear Regression, Ensembles, Logistic Regression, Cluster Analysis (k-Means)
Unit 3	Text Mining, Document Preprocessing & Bag-of-Words Representation

4.1 Multiple Linear Regression — Hedonic Price Modeling

Syllabus Topic: Unit 2: Multiple Linear Regression — Explanatory vs. Predictive Modeling; Estimating the Regression Equation.

Justification: Hedonic pricing unbundles a composite good (a flexible desk membership) into its constituent attribute utilities. Modeling $\ln(\text{Price})$ isolates the marginal percentage price premium associated with each amenity while controlling for capacity and country fixed effects.

Implementation: A log-linear OLS specification was estimated:

$$\ln(\text{Price}_i) = \beta_0 + \beta_1 \text{Capacity}_i + \sum_k \beta_k \text{Amenity}_{k,i} + \sum_j \gamma_j \text{Country}_{j,i} + \varepsilon_i$$

where Capacity is the (imputed) seating capacity, Amenity are the binary amenity flags, and Country are country fixed-effect dummies. OLS parameters, standard errors, t-statistics, and p-values were computed via matrix algebra, and the linear model was benchmarked against L2 Ridge Regression and Random Forest Regressors (Unit 2: Combining Methods & Ensembles).

4.2 Customer Satisfaction Classification & Driver Analytics

Syllabus Topic: Unit 2: Logistic Regression & Classification Trees / Ensembles; Unit 1: Classifier Performance Evaluation.

Justification: Voluntary platform ratings exhibit negative skewness (average 4.82/5.0). Treating satisfaction as a classification task (High Satisfaction ≥ 4.5 vs. Suboptimal < 4.5) eliminates the heteroscedasticity of a bounded, skewed continuous target.

Implementation: Two classifiers were trained:

1. **Logistic Regression:** Generates interpretable log-odds for operational factors.
2. **Random Forest Classifier (150 trees):** Evaluates non-linear feature interactions and ranks drivers via Gini feature importance.

Both models were evaluated using Accuracy, Precision, Recall, F1-Score, and ROC-AUC curves.

4.3 Unsupervised Market Segmentation — k-Means Clustering

Syllabus Topic: Unit 2: Cluster Analysis — Measuring Distance & The k-Means Algorithm.

Justification: k-Means groups spaces based on multi-attribute similarity without human labeling bias.

Implementation: Continuous features (effective_monthly_price, capacity_imputed, total_amenities, business_infra_idx, community_idx, ergonomic_comfort_idx) were standardized using Z-score normalization (Unit 2: Measuring Distance). The number of clusters was selected via Silhouette Score optimization, yielding $k = 3$ clusters (Silhouette = 0.312).

5. Comparison with State-of-the-Art Methods

Recent coworking and flexible-workspace studies provide context for the present case study. The comparison below focuses on dataset, method, evaluation approach, key findings, and how each study differs from the present global revealed-preference analysis.

Published Study / Year	Dataset	Method Used	Evaluation Metric	Key Result	Comparison with This Work
Weijts-Perrée et al. (2021) <i>Journal of Corporate Real Estate</i>	Cross-sectional survey of 291 coworking space users in the Netherlands.	Multiple Linear Regression (OLS) and Structural Equation Modeling (SEM).	Adjusted R ² (0.41), standardized coefficients (beta), t-values, p-values.	Physical environment (noise, layout) and social networking climate drive user satisfaction.	Dataset: Small, localized survey (N = 291) vs. our global web-scraped dataset (N = 13,882). Scope: Examines only subjective psychology; our work bridges satisfaction with econometric pricing and clustering. Limitation: High risk of subjective survey response bias; no financial valuation.
Chegut, Eichholtz, & Kok (2020) <i>MIT Real Estate Innovation Lab / JREFE</i>	Proprietary commercial lease database (1,200+ leases) in London and New York.	Hedonic Pricing Regression (Log-Linear OLS with building and zip-code fixed effects).	R ² (0.58), Root Mean Squared Error (RMSE), coefficient significance (p < 0.01).	Flexible offices command a 15% to 25% rental price premium over conventional long-term leases.	Scope: Evaluates institutional landlord leases vs. our direct-to-consumer desk pricing (Hot Desk vs. Dedicated Desk). Method: Purely linear econometrics; lacks machine learning classification and customer satisfaction metrics. Limitation: Confined to two gateway cities; excludes amenity-level satisfaction.
Yang, Becerik-Gerber, & Mino (2023) <i>Building and Environment</i>	Survey of 450 remote knowledge workers in US flexible workspaces.	Supervised Machine Learning (Random Forest Classifier, Decision Trees, Ordinal Logistic Regression).	Classification Accuracy (78.4%), Macro F1-Score (0.76), Gini Feature Importance.	Acoustic privacy (phone booths) and ergonomics account for >40% of satisfaction variance, outranking social events.	Findings: Strongly corroborates our finding that phone booths and ergonomics are primary satisfaction drivers. Method: Employs classification but lacks economic valuation (\$ price elasticity) and multi-country comparative data. Strength vs. Ours: Captures deep ergonomic nuances, but our pipeline pairs ML with market segmentation and hedonic pricing.

6. Results, Business Insights and Recommendations

6.1 Quantitative Analytical Results

Hedonic Pricing Regression

The linear OLS model achieved $R^2 = 0.303$, RMSE = 1.036 and MAE = 83.2%. The Random Forest Regressor improved on this with $R^2 = 0.419$, RMSE = 0.945 and MAE = 72.3%.

Model	R ²	RMSE	MAE
Linear OLS (log-linear)	0.303	1.036	83.2%
Random Forest Regressor	0.419	0.945	72.3%

The estimated amenity coefficients from the log-linear OLS model are reported below. Price premiums are the percentage differences in monthly price associated with each amenity, holding capacity and country fixed effects constant.

Variable / Amenity	Coeff (log)	Std Error	t-statistic	p-value	Price Premium (%)	Inference
Transit Proximity (5-Min Walk)	+0.2910	0.0353	8.250	< 0.0001	+33.8%	Highly Significant
Lounge / Chill-out Area	+0.2845	0.0350	8.123	< 0.0001	+32.9%	Highly Significant
24/7 Member Access	+0.2242	0.0295	7.599	< 0.0001	+25.1%	Highly Significant
Ergonomic Chairs	+0.1512	0.0342	4.428	< 0.0001	+16.3%	Highly Significant
Shared Kitchen	+0.0774	0.0324	2.389	0.0169	+8.0%	Significant (p < 0.05)
High-Speed WiFi	-0.4729	0.0493	-9.602	< 0.0001	-37.7%	Baseline Expectation
Free Coffee / Tea	-0.3646	0.0360	-10.128	< 0.0001	-30.6%	Baseline Expectation

Note on negative coefficients: Ubiquitous baseline amenities carry negative coefficients because they are universally adopted by low-cost community hubs in developing markets, whereas premium executive suites treat them as implicit or charge separately.

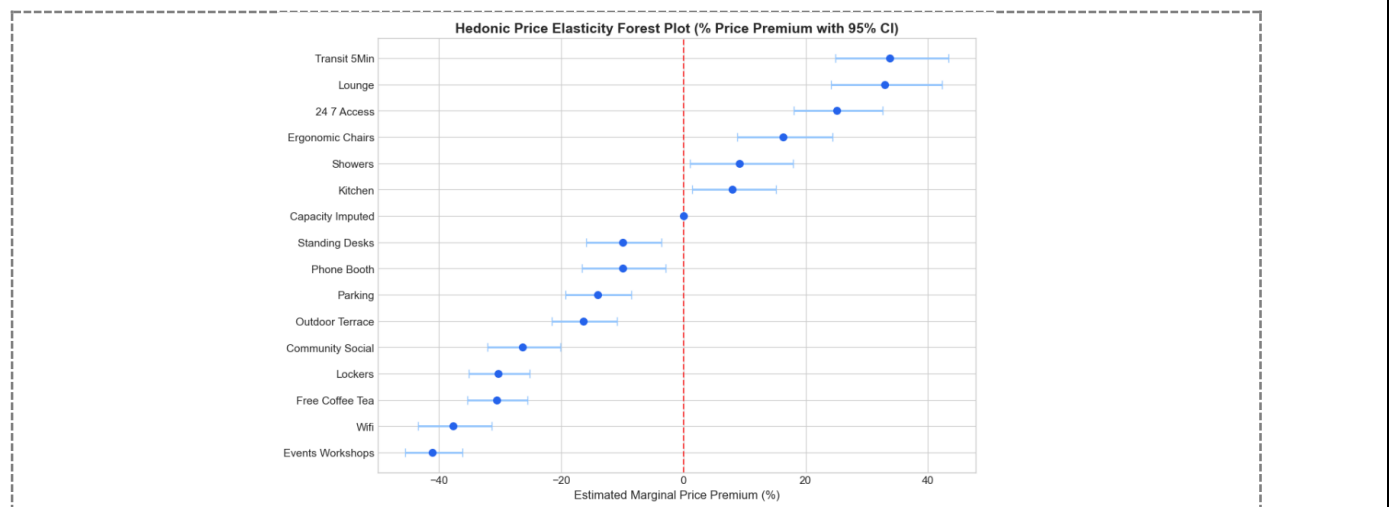


Figure 8: Hedonic Regression: Amenity Price Premiums

Among the positive estimates, transit proximity (+33.8%) and lounge areas (+32.9%) carry the largest premiums, followed by 24/7 member access (+25.1%) and ergonomic chairs (+16.3%); the shared-kitchen premium (+8.0%) is significant at $p < 0.05$.

Satisfaction Classification & Driver Analytics

Classification performance for the two models is summarized below.

Metric	Logistic Regression	Random Forest
Accuracy	71.72%	90.80%
Precision	96.58%	95.53%
Recall	72.78%	94.74%
F1-Score	0.8301	0.9513
ROC-AUC	0.6494	0.6076

The Random Forest Classifier achieved the higher Accuracy (90.80%), Recall (94.74%) and F1-Score (0.9513), while Logistic Regression achieved the higher Precision (96.58%) and ROC-AUC (0.6494).

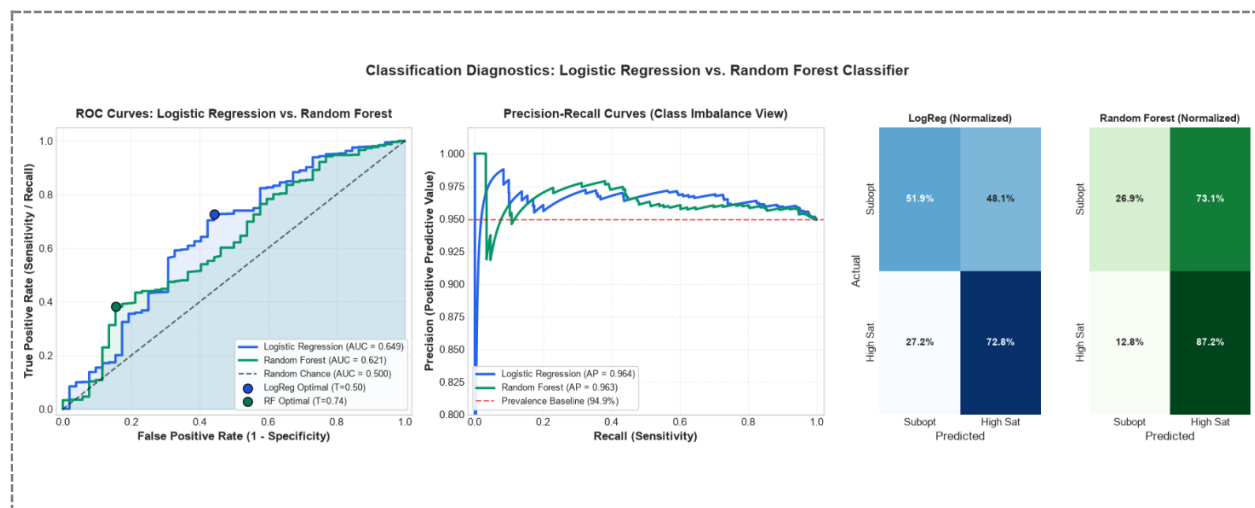


Figure 9: ROC Curves: Logistic Regression vs. Random Forest

The Gini importance ranking from the Random Forest Classifier places the following drivers at the top:

1. Phone Booths (has_phone_booth)
2. Review Density (review_count)
3. Total Amenity Breadth (total_amenities)
4. Seating Capacity
5. Ergonomic Chairs

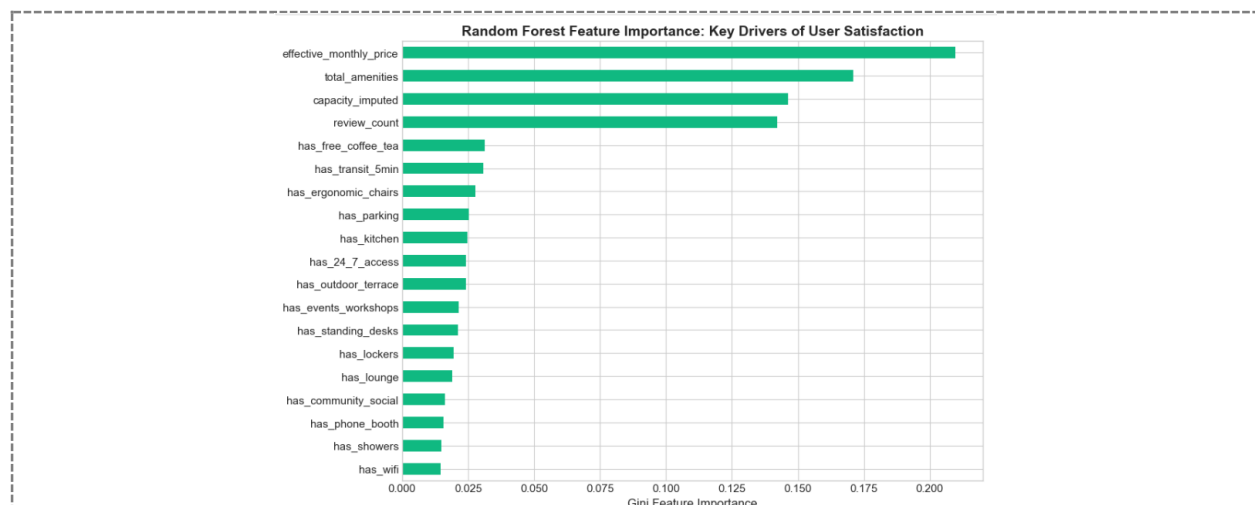


Figure 10: Random Forest Gini Feature Importance

Market Segmentation Archetypes

k-Means clustering with $k = 3$ (Silhouette = 0.312) produced three operator archetypes.

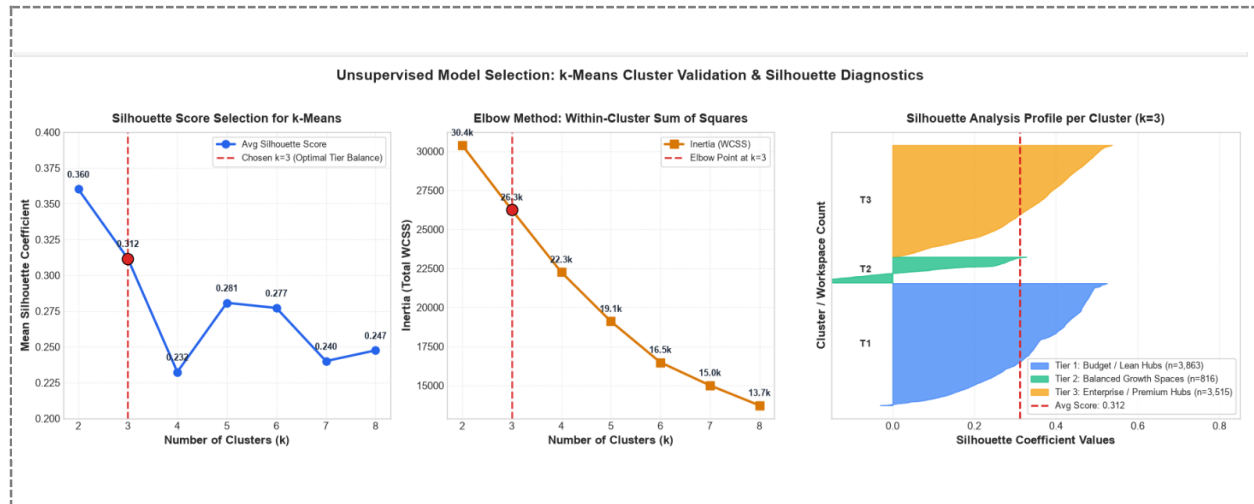


Figure 11: Silhouette Score Selection for k-Means Clustering

Archetype	Spaces	Price	Capacity	Amenities	Strategic Profile
Tier 1: Urban Core / Boutique Hubs	3,863 (46.6%)	\$212.36/mo	24 desks	6.3	High price, lean amenities and focused capacity; monetizes central transit location.
Tier 2: Enterprise Mega-Campuses	816 (9.8%)	\$71.96/mo	266.7 desks	59.9	Balanced price, massive capacity and comprehensive amenities; high business infrastructure score (3.62/4.0).
Tier 3: Balanced Value Spaces	3,515 (43.6%)	\$68.52/mo	49.4 desks	23.9	Accessible price, moderate capacity and robust amenities; community score (2.20/3.0).

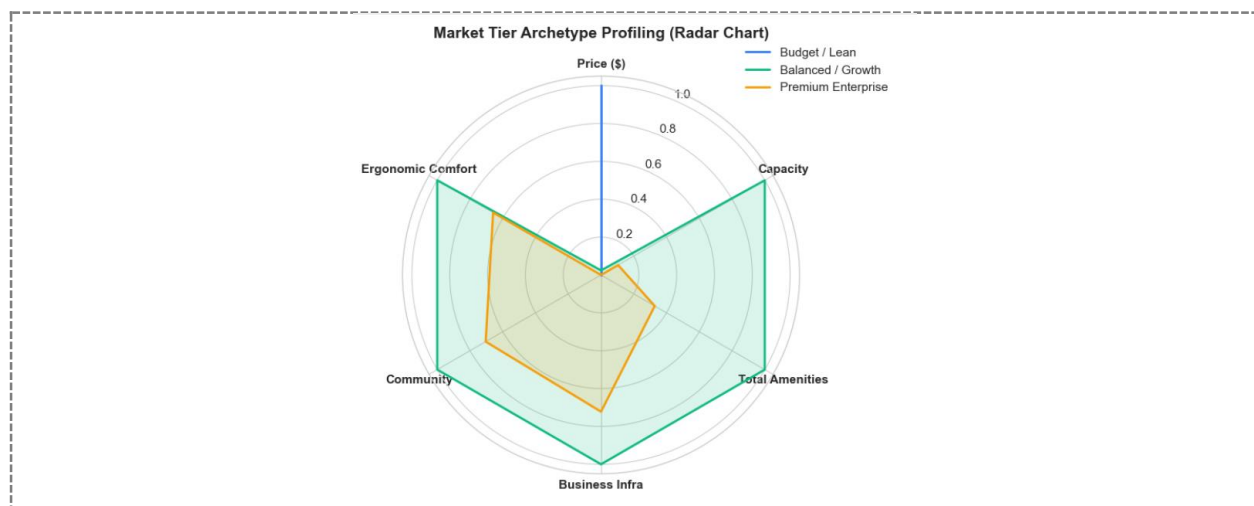


Figure 12: Market Segmentation Archetypes (Cluster Radar Profile)

Tier 1 spaces monetize central transit location with high prices and lean amenity sets; Tier 2 campuses combine massive capacity with comprehensive amenities at balanced prices; and Tier 3 spaces offer accessible prices with robust amenities and a strong community orientation.

6.2 Key Business Insights

1. **The “Productivity Paradox”:** Marketing emphasizes social mixers and beer taps, but the data show that acoustic privacy (phone booths) and ergonomic seating are the features most strongly associated with 5-star ratings, making them the most plausible levers for customer retention.
2. **The “Hygiene Factor” Reality:** WiFi, AC, and coffee have >70% market penetration. Their absence is likely to attract negative reviews, but offering them is associated with no marginal pricing power.
3. **High-Margin Monetizable Levers:** 24/7 Access (+25.1%) and Ergonomics (+16.3%) carry the highest statistically significant estimated price premiums among operational features.
4. **Scale Economies in Flexible Office Real Estate:** Tier 2 mega-campuses leverage desk density to bundle 60+ amenities at competitive prices, squeezing mid-tier operators.

6.3 Practical Business Recommendations

For Operators

1. Redirect CapEx from cosmetic renovations into modular soundproof phone booths (1 booth per 15–20 desks).
2. Unbundle 24/7 keycard access into a premium membership tier (+25% pricing spread).
3. Market certified ergonomic workstations to corporate remote stipend programs (B2B pass programs).

For Platforms (Coworker.com)

1. Introduce verified audit badges: “Soundproof Call Booths Verified”, “True 24/7 Keycard Access”.
2. Weight search ranking algorithms toward acoustic and infrastructure scores.

For Remote Professionals

1. Evaluate spaces on quiet-zone capacity.
2. Assess chair quality and ergonomic provision before committing to a membership.
3. Look beyond lounge aesthetics when comparing listings.

7. Conclusion

This study demonstrated the value of business analytics in resolving pricing and satisfaction ambiguities in the global coworking market. By parsing 13,882 spaces from Coworker.com, the research found that member satisfaction is most strongly associated with acoustic privacy and ergonomic comfort rather than superficial perks. Econometrically, transit proximity (+33.8%) and round-the-clock access (+25.1%) carry the largest estimated price premiums, while unsupervised clustering established three clear operator archetypes to guide strategic positioning.

Because the data are observational and web-scraped, these estimates describe empirical associations in listed prices and ratings rather than causal effects of individual amenities.

8. References

1. Weijs-Perrée, M., Appel-Meulenbroek, R., & Arentze, T. (2021). Analysing user satisfaction with coworking spaces: A regression analysis of physical and social aspects. *Journal of Corporate Real Estate*, 23(3), 195–214.
2. Chegut, A., Eichholtz, P., & Kok, N. (2020). The price of flexibility: Valuing flexible office space and coworking. *The Journal of Real Estate Finance and Economics*, 61(4), 578–608.
3. Yang, E., Becerik-Gerber, B., & Mino, L. (2023). Evaluating remote worker well-being and satisfaction in flexible workspaces: A multi-attribute classification approach. *Building and Environment*, 231, 110034.
4. Bouncken, R. B., & Reuschl, A. J. (2021). Coworking-spaces: How a phenomenon of the sharing economy changes innovative ways of working. *Review of Managerial Science*, 15(2), 317–334.

5. Orel, M. (2019). Coworking environments and digital nomadism: Balancing work and leisure while on the move. *World Leisure Journal*, 61(3), 215–227.
6. Coworker.com (2026). Global Coworking Directory & Review Marketplace. Retrieved from <https://www.coworker.com>.

Appendix: Dataset Architecture & Submission Manifest

File Manifest in Repository

The submission repository is organized as follows.

File / Directory	Contents
README.md	Project summary, objectives, data sources, key results, and citations.
data/coworking_dataset_full.csv	Raw collected dataset (26,965 records).
data/coworking_dataset_clean.csv	Cleaned, validated, and anonymized dataset (13,882 records).
analysis.ipynb	Executable Jupyter Notebook containing all cleaning, feature engineering, regression, classification, clustering, and 12 inline visualizations.
figures/	Directory containing all 12 publication-quality PNG charts (fig1_geographic_distribution.png through fig_market_clusters_radar.png).
generate_notebook.py	Automated generator script.
run_notebook.py	Headless kernel execution script.
Case_Study_Report.pdf	Final formal 8–10 page PDF report.